

HOSPITAL EMERGENCY QUEUE MANAGEMENT SYSTEM IMPLEMENTATION REPORT

Project Overview

Implementation of a full-stack emergency queue system.Implementation of a full-stack emergency queue system.Implementation of a full-stack emergency queue system.Implementation of a full-stack emergency queue system.

Backend Design

Express.js REST APIs with heap-based priority queue.Express.js REST APIs with heap-based priority queue.Express.js REST APIs with heap-based priority queue.Express.js REST APIs with heap-based priority queue.

Priority Queue Implementation

Patients are ordered by ESI severity and arrival timestamp.Patients are ordered by ESI severity and arrival timestamp.Patients are ordered by ESI severity and arrival timestamp.Patients are ordered by ESI severity and arrival timestamp.

API Endpoints

POST /patients, GET /patients/next, PUT /patients/:id and related endpoints.POST /patients, GET /patients/next, PUT /patients/:id and related endpoints.POST /patients, GET /patients/next, PUT /patients/:id and related endpoints.POST /patients, GET /patients/next, PUT /patients/:id and related endpoints.

Frontend Design

React components for triage form, dashboard, and doctor view.React components for triage form, dashboard, and doctor view.React components for triage form, dashboard, and doctor view.React components for triage form, dashboard, and doctor view.

Real-Time Communication

Socket-based updates for dashboards.Socket-based updates for dashboards.Socket-based updates for dashboards.Socket-based updates for dashboards.

Database Schema

PostgreSQL schema for patients, doctors, and triage levels.PostgreSQL schema for patients, doctors, and triage levels.PostgreSQL schema for patients, doctors, and triage levels.PostgreSQL schema for patients, doctors, and triage levels.

Deployment

Docker containers and Kubernetes orchestration.Docker containers and Kubernetes orchestration.Docker containers and Kubernetes orchestration.Docker containers and Kubernetes orchestration.

Testing

Simulation scripts and automated tests.Simulation scripts and automated tests.Simulation scripts and automated tests.Simulation scripts and automated tests.

Execution Results

Backend and frontend services run successfully.Backend and frontend services run successfully.Backend and frontend services run successfully.Backend and frontend services run successfully.

Source Code Explanation

Detailed explanation of heap operations, REST APIs, and UI components.Detailed explanation of heap operations, REST APIs, and UI components.Detailed explanation of heap operations, REST APIs, and UI components.Detailed explanation of heap operations, REST APIs, and UI components.

Conclusion

Successful implementation of a scalable hospital emergency queue platform.Successful implementation of a scalable hospital emergency queue platform.Successful implementation of a scalable hospital emergency queue platform.Successful implementation of a scalable hospital emergency queue platform.